Financial Market Analysis

Portfolio Optimization, Risk Management & Monte Carlo Simulation

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Agenda

1. Overview & Data
2. Price & Return Analysis
3. Portfolio Optimization
4. Risk Management
5. Monte Carlo Simulation
6. Regime Analysis
7. Conclusions & Future Work

Scope & Asset Universe

Objectives

- Analyze 8 assets across 5 sectors over 10+ years
- Construct **Markowitz efficient frontier**
- Measure risk with **VaR, CVaR, Sharpe, Sortino**
- Project outcomes via **Monte Carlo simulation**
- Detect market regimes (bull / bear / high-vol)

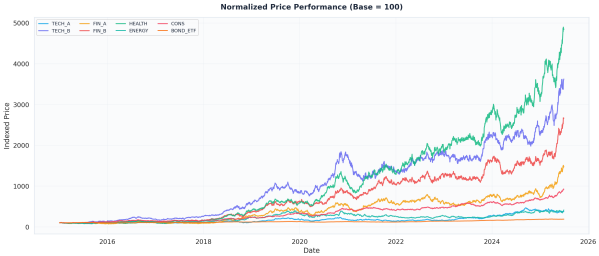
Asset Universe

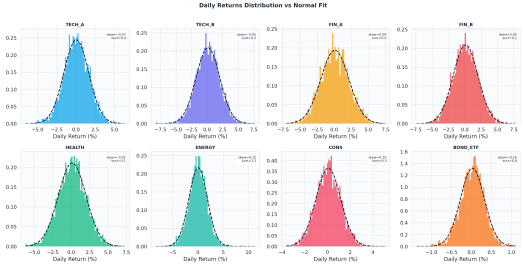
Ticker	Sector	Profile
TECH_A/B	Technology	High growth/vol
FIN_A/B	Financials	Market beta
HEALTH	Healthcare	Defensive
ENERGY	Energy	Commodity-linke
CONS	Consumer	Stable
BOND_ETF	Fixed Income	Low risk

Data

2,737 trading days (Jan 2015 – Jun 2025). Daily prices, returns, volume, and market regime

Normalized Performance & Return Distributions

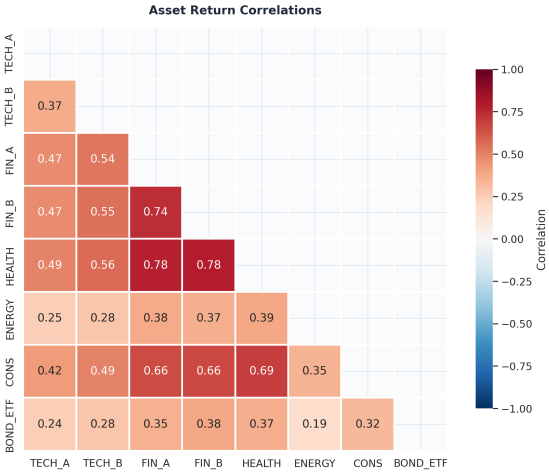


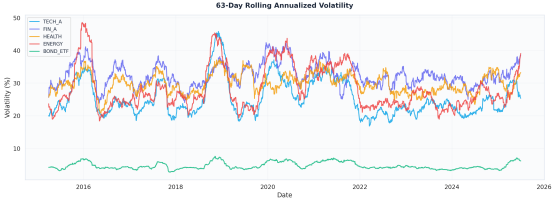


Key Observations

- Technology and healthcare assets delivered the highest total returns
- All return distributions exhibit **fat tails and excess kurtosis** — normal assumptions underestimate tail risk
- Jarque-Bera tests reject normality for every asset ($p < 0.001$)

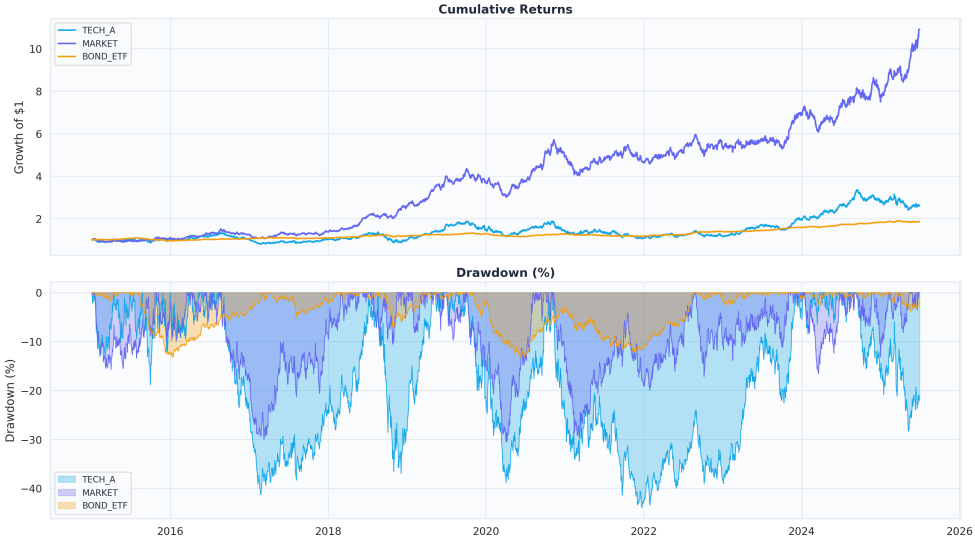
Correlation & Rolling Volatility



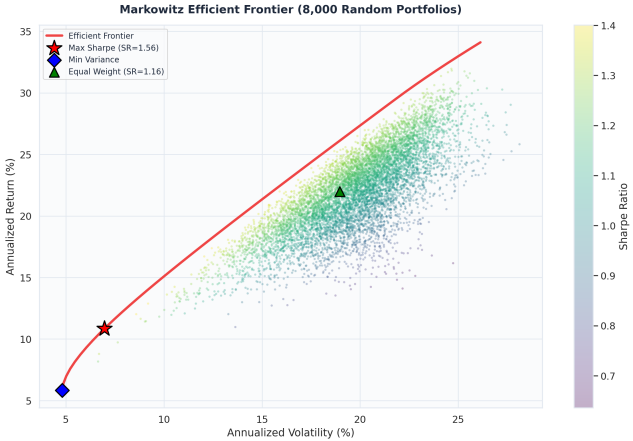


Diversification Opportunity

Drawdown Analysis



Markowitz Efficient Frontier



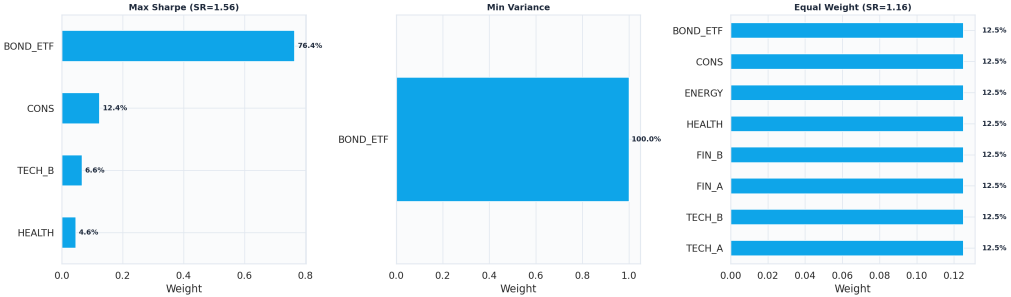
Optimal Portfolios

	Ret %	Vol %	SR
Max Sharpe	10.8	7.0	**1.56**
Min Var	5.1	4.9	1.04
Equal Wt	9.9	11.5	0.86

Insight

The Max Sharpe portfolio achieves **1.56 Sharpe** by heavily allocating to low-volatility assets. Naive equal-weighting delivers similar return but with **60% more risk**.

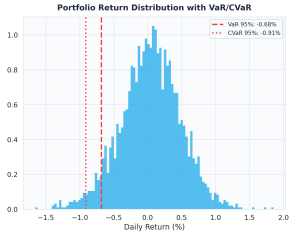
Portfolio Allocations

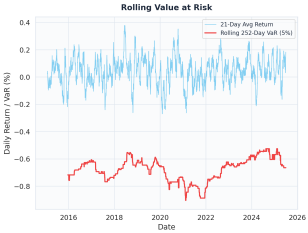


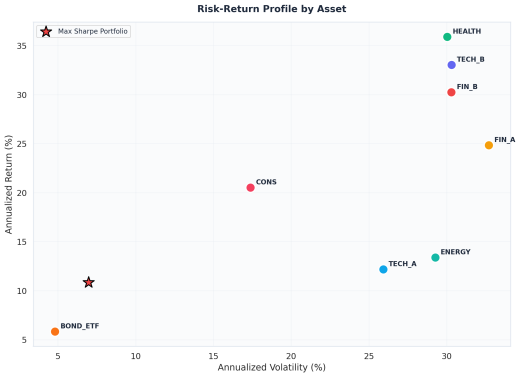
Allocation Interpretation

The Max Sharpe portfolio favors BOND_ETF, CONS, and HEALTH — assets with the best **risk-adjusted returns**. The optimizer avoids high-volatility assets (ENERGY, TECH_B) despite their higher raw returns.

Value at Risk & Risk-Return Profile



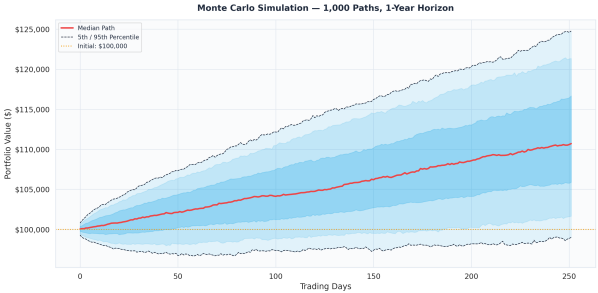


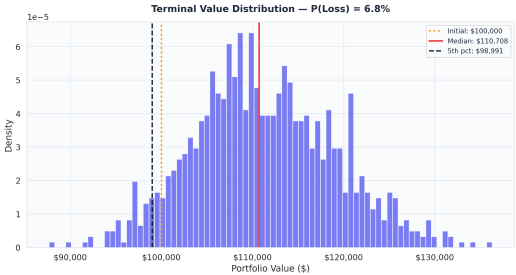


Risk Metrics (Max Sharpe Portfolio)

Metric	Value	Metric	Value
Daily VaR (95%)	–0.68%	Sharpe Ratio	1.56

1-Year Portfolio Projection



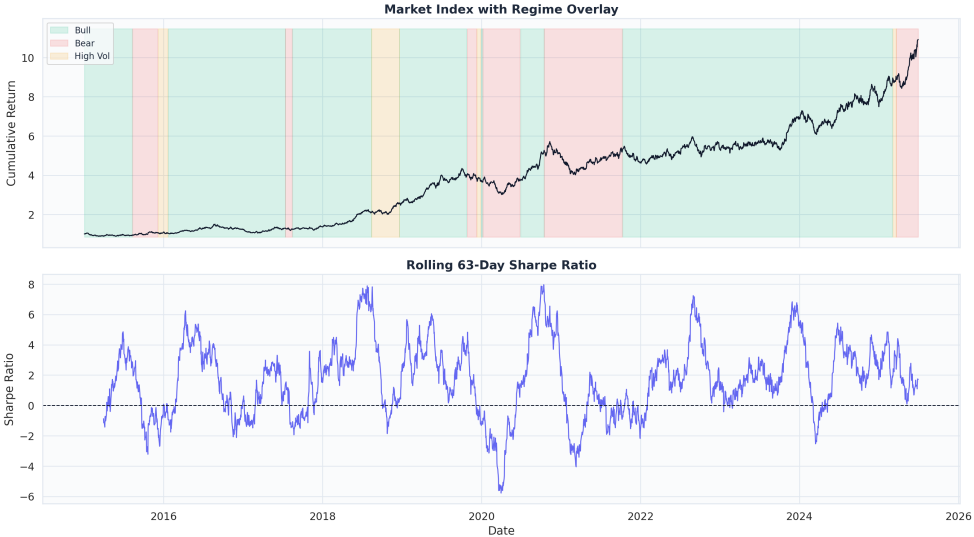


Simulation Results ($100K initial, 1,000 paths)

Median outcome	$110,708	P(loss)	6.8%
5th percentile	$98,991	95th pct	$124,771

Interpretation

Market Regimes & Rolling Sharpe



Summary & Recommendations

Key Findings

1. Max Sharpe portfolio: **SR = 1.56** with 7% vol
2. Fat tails everywhere — **normal VaR underestimates** risk
3. Fixed income is critical for diversification
4. Monte Carlo: **93% chance** of profit over 1 year
5. Regime shifts materially impact performance

Future Work

1. GARCH / EGARCH volatility modeling
2. Black-Litterman with subjective views
3. Fama-French factor attribution
4. Options-based tail hedging
5. RL for dynamic allocation
6. Regime-switching models (HMM)

Thank You

Questions & Discussion

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